

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 5: Arithmetic Progressions

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sum of the first n terms

1. In an A.P., if the first term $a = 7$, n th term $a_n = 84$ and the sum of first n terms $S_n = \frac{2093}{2}$, then n is equal to: (a) 22 (b) 24 (c) 23 (d) 26 **CBSE 2024, 1 mark**

SOLUTION

$$S_n = \frac{n}{2}(a + a_n) = \frac{n}{2}(7 + 84) = \frac{91n}{2}$$

$$\frac{91n}{2} = \frac{2093}{2} \text{ gives } n = 23$$

Answer: (c)

2. The sum of first and eighth terms of an A.P. is 32 and their product is 60. Find the first term and common difference of the A.P. Hence, also find the sum of its first 20 terms. **CBSE 2024, 5 marks**

SOLUTION

$$a + (a + 7d) = 32, \text{ so } a + 7d = 32 - a.$$

$$\text{Product: } a(32 - a) = 60, \text{ so } a^2 - 32a + 60 = 0 \text{ and } (a - 2)(a - 30) = 0. \text{ Hence } a = 2 \text{ or } a = 30.$$

$$\text{If } a = 2: 7d = 28, \text{ so } d = 4. \text{ Then } S_{20} = \frac{20}{2}[2(2) + 19(4)] = 10 \times 80 = 800.$$

$$\text{If } a = 30: 7d = -28, \text{ so } d = -4. \text{ Then } S_{20} = \frac{20}{2}[2(30) + 19(-4)] = 10 \times (-16) = -160.$$

Answer: $a = 2, d = 4, S_{20} = 800$; or $a = 30, d = -4, S_{20} = -160$.

3. In an A.P. of 40 terms, the sum of first 9 terms is 153 and the sum of last 6 terms is 687. Determine the first term and common difference of the A.P. Also, find the sum of all the terms of the A.P. **CBSE 2024, 5 marks**

SOLUTION

$$S_9 = \frac{9}{2}[2a + 8d] = 153, \text{ so } a + 4d = 17.$$

$$\text{The last 6 terms are } a_{35} \text{ to } a_{40}. \text{ Their sum} = \frac{6}{2}(a_{35} + a_{40}) = 3(2a + 73d) = 687, \text{ so } 2a + 73d = 229.$$

$$\text{Substituting } a = 17 - 4d: 34 - 8d + 73d = 229, \text{ so } 65d = 195 \text{ and } d = 3. \text{ Then } a = 5.$$

$$S_{40} = \frac{40}{2}[2(5) + 39(3)] = 20 \times 127 = 2540$$

Answer: $a = 5, d = 3$, and the sum of all 40 terms is 2540.